

# Intuition & Rigor

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The scaffolding we burn building the truth.

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What remains..



# The scaffolding we burn building the truth.

Textbooks present a sanitized history: neat derivations that end at the theorem. They rarely show how the question itself was born. Physics is not a tidy march from hypothesis to proof; it's a workshop of approximations and educated guesses. Rigor arrives later, not as the opposite of intuition, rather as its adjudicator. This book reconstructs the missing half of your training. It will not replace formal theory; it will teach you how to frame problems, diagnose which tools matter, and build the temporary scaffolding that leads to insight. Once the answer stands, you can formalize it and then burn the scaffolding. This book is the ash.

*Vojtěch Kuchař*



# CHAPTER I.

# IMMUTABLE

*Constraints precede dynamics.*

Physics does not care about your coordinate system. Before you write down a single equation, you must ask: what stays the same when I change how I look?

Galileo noticed this first. Experiments on a moving ship behave as if the ship were still. Noether made it rigorous: every symmetry hides a conservation law. Dynamics is not the starting point, it is the consequence.

This chapter is about the constraints that exist before anything moves. You will not solve problems by summing forces; you will solve them by finding the transformation under which the system cannot tell the difference.

## **The Method of Scales.**

No characteristic length means no preferred ruler. The answer must be a power law.

## **The Method of Mirrors.**

A plane of symmetry cuts your work in half. One side determines the other.

## **The Method of Frames.**

There is no absolute observer. Jump to the frame where the motion simplifies, where velocity or acceleration vanishes.

## **The Method of Gauges.**

Potentials are fictions. Add whatever terms you like to them, as long as the physics stays unchanged. Use that freedom to decouple your equations.

## **The Method of Maps.**

A spring is a capacitor. A mass is an inductor. If you have solved one problem, you have solved every problem with the same structure.

## **The Method of Duals.**

Sometimes strong coupling is weak coupling in disguise. Invert the parameter, and the intractable becomes trivial.

The goal is always the same: turn differential equations into algebra.

## §1. The Method of Scales

**Problem 1.** *In 1945, the physicist G.I. Taylor sought to determine the energy yield of the first atomic explosion. The data available to him was sparse: a series of photographs showing the radius of the fireball  $R$  expanding over time  $t$ , and the knowledge of the ambient air density  $\varrho$ .*

**Solution.** In violent events like this, the physics is fully governed by the event itself. Initial conditions do not play a role, the explosion must be guided solely by the parameters of the fireball. Let us seek a relationship of the form

$$R = f(E, \varrho, t). \quad (1.1)$$

Assuming

$$R \propto E^a \varrho^b t^c, \quad (1.2)$$

dimensional analysis yields

$$L = (ML^2T^{-2})^a (ML^{-3})^b (T)^c \implies \begin{aligned} a + b &= 0, \\ -2a + c &= 0, \\ 2a - 3b &= 1, \end{aligned} \quad (1.3)$$

which implies

$$R = C \left( \frac{Et^2}{\varrho} \right)^{1/5}, \quad (1.4)$$

where  $C$  is an unknown constant of proportionality. Dimensionless constants are usually geometry-related, rarely being more than one order away from unity. Given the scaling of  $1/5$ , it is safest to assume  $C \approx 1$ . This leads to

$$R = \left( \frac{Et^2}{\varrho} \right)^{1/5}. \quad (1.5)$$

**Note.** The reason why dimensional analysis leads us to the goal here is the violence of the blast itself.

**Proof (after L. I. Sedov, G.I. Taylor, and J. von Neumann).** We model the medium as an ideal polytropic gas with adiabatic index  $\gamma$ . The motion is governed by the system of Euler equations for the conservation of mass, momentum, and entropy in spherical coordinates....

□

## §2. The Method of Mirrors

## §3. The Method of Frames

## §4. The Method of Gauges

## §5. The Method of Maps

## §6. The Method of Duals



## CHAPTER II.

# INEVITABLE

§7. The Method of Paths

§8. The Method of Trials

§9. The Method of Flows

§10. The Method of Loops

§11. The Method of Catastrophes

## §12. The Method of Histories

CHAPTER III.

GEOMETRIC

§13. The Method of Lattices

§14. The Method of Modes

§15. The Method of Phases

§16. The Method of Ghosts

§17. The Method of Knots



CHAPTER IV.

**EMERGENT**

§18. The Method of Blurs

§19. The Method of Crowds

§20. The Method of Noise

§21. The Method of Grains

§22. The Method of Vacuua



CHAPTER V.

HOLOGRAPHIC

§23. The Method of Shadows

§24. The Method of Demons

§25. The Method of Codes

§26. The Method of Holograms

§27. The Method of Amplitudes



# WHAT REMAINS

**Some bold text.**

$$\left(\left[\left[\frac{x}{n}\right]\right]\right) \leq \left\{\left[\frac{x}{n}\right]\right\}$$
$$\prod_{p \leq x} p \leq \sqrt[4]{avbc}$$

$a, b, c, \alpha, \beta, \gamma, \varphi, \phi, \psi, \Psi, \varrho, \partial$ . THIS IS SMALLCAPS TEXT